HDOC Day-49

AIM

To write a C program to find the majority element in a variable-length integer array using pointers and a user-defined function named **majority_element()**.

PROBLEM STATEMENT

Read the size and elements of an integer array **nums[]**. Pass the array to **majority_element()** using its address. Count the occurrences of each distinct element and display the element that occurs strictly more than floor(size / 2) times. Display -1 if no majority element exists.

ALGORITHM

1. Start the program.
2. Read the size of the array.
3. Read all array elements.
4. Pass the address of the array and its size to majority_element().
5. For each element, count how many times it occurs using a nested loop.
6. If its count is greater than size / 2, display it as the majority element and stop checking.
7. If no element satisfies the condition, display -1.
8. Stop the program.

C PROGRAM

```
#include <stdio.h>

void majority_element(int *nums, int size)
{
    int i, j, count;

    for(i = 0; i < size; i++)
    {
        count = 0;

        for(j = 0; j < size; j++)
        {
            if(*(nums + i) == *(nums + j))
            {
                count++;
            }
        }

        if(count > size / 2)
        {
            printf("Majority element: %d\n", *(nums + i));
            return;
        }
    }

    printf("-1\n");
}

int main()
```

```
{
    int size, i;

    printf("Enter size of array: ");
    scanf("%d", &size);

    int nums[size];

    printf("Enter array elements: ");
    for(i = 0; i < size; i++)
    {
        scanf("%d", &nums[i]);
    }

    majority_element(nums, size);

    return 0;
}
```

TEST CASE TABLE

Test Case	Input Array	Expected Output
1	3 3 2 3 3	Majority element: 3
2	2 2 1 1 1 2 2	Majority element: 2
3	1 2 3 2 3 3	-1

OUTPUT

Test Case 1

```
Enter size of array: 5
Enter array elements: 3 3 2 3 3
Majority element: 3
```

Test Case 2

```
Enter size of array: 7
Enter array elements: 2 2 1 1 1 2 2
Majority element: 2
```

Test Case 3

```
Enter size of array: 6
Enter array elements: 1 2 3 2 3 3
-1
```

RESULT

Thus, the C program to find and display the majority element of an integer array using pointers was implemented successfully.